

Building and Comparing Two Number Patterns

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|----------|----------|-----------|-------|
| 1. 10 | 4. 24 | 7. 11, 22 | 10. — |
| 2. 30 | 5. 26 | 8. 15 | |
| 3. 3, 30 | 6. 52, 2 | 9. 10 | |
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Curriculum

Level: Grades 4–5 Mathematics

5.G.A.2 – *problems 4, 9*

5.OA.B.3 – *problems 1, 2, 3, 5, 6, 7, 8, 10*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

8. *If they got 18: doubled the first pattern's fourth term because the step was doubled*

1. Pattern A: start at 0, add 2 each time.

Solution: Write the starting number as term 1, then add the step once for every move to the next term: 0, 2, 4, 6, 8, 10. Getting from term 1 to term 6 is five steps of 2, which is $0 + 2 \times 5$. 10

Common slip to watch for: adding 2 six times instead of five and answering 12. Term 1 is the start, before any adding.

2. Pattern B: start at 0, add 6 each time.

Solution: Same reasoning with a bigger step: 0, 6, 12, 18, 24, 30. From term 1 to term 6 is five steps of 6, so $0 + 6 \times 5$. 30

3. Comparing A and B term by term.

Lined up, the two patterns look like this:

Pattern A	0	2	4	6	8	10
Pattern B	0	6	12	18	24	30

(a) Divide the fourth terms, B's over A's: $18 \div 6$. 3

The same answer comes from any matching pair: $6 \div 2 = 3$, $12 \div 4 = 3$, $24 \div 8 = 3$.

(b) Because every B term is 3 times the matching A term, and A's sixth term is 10, B's sixth term is 3×10 . 30

That matches the sixth term found by adding in problem 2, which is the check. The reason the relationship is exactly "times 3" here is that B's step, 6, is 3 times A's step, 2, *and* both patterns start at 0.

4. Ordered pairs from A and B.

(a) (Open response — judge by reading.) Pair up corresponding terms, A first: (0, 0), (2, 6), (4, 12), (6, 18). Plotted on the grid, the four points climb from the origin in a straight line, going 2 across and 6 up each time.

(b) Every pair's second number is 3 times its first number, so with 8 as the first number the second number is 3×8 . 24

Look for: pairs written first number from Pattern A, points plotted at the correct cross-ings (not shifted by one square), and a straight-line arrangement noticed or at least not contradicted.

5. Ana's jar: 5 dollars in week 1, add 3 dollars a week.

Solution: Weeks 1 to 4: 5, 8, 11, 14 dollars. Week 8 is seven steps after week 1, so it is $5 + 3 \times 7 = 5 + 21$ dollars. 26

Counting the steps rather than the weeks is the whole trick: eight weeks, seven additions.

6. Ben's jar: 10 dollars in week 1, add 6 dollars a week.

(a) Week 8 is again seven steps after week 1: $10 + 6 \times 7 = 10 + 42$ dollars. 52

(b) Divide Ben's week-8 total by Ana's: $52 \div 26$. 2

Ben's jar holds twice Ana's in every week, because Ben started with twice as much *and* adds twice as much each week. Both halves are needed.

7. Means of the two savers.

(a) Ana: $5 + 8 + 11 + 14 + 17 = 55$ dollars in total, shared over five weeks: $55 \div 5$. 11

(b) Ben: $10 + 16 + 22 + 28 + 34 = 110$ dollars in total, over the same five weeks: $110 \div 5$. 22

(c) Yes: 22 is twice 11. Every one of Ben's weekly totals is twice Ana's matching total, so the two totals added up are in the same ratio, and dividing both by 5 keeps it. A relationship that holds term by term is inherited by the mean.

8. Patterns C and D — checking Mia’s claim.

Solution: Write both patterns out to term 4.

Pattern C (start 3, add 2)	3	5	7	9
Pattern D (start 3, add 4)	3	7	11	15

Pattern D’s fourth term is $3 + 4 \times 3 = 3 + 12$.

15

Mia answered 18, which is twice Pattern C’s fourth term of 9. What her rule missed: doubling the step does not double the terms unless the starting number doubles too. Here both patterns start at the same 3, so that shared 3 is never doubled — Pattern D’s term is always Pattern C’s term plus the extra steps, 2 more for each step taken, and at term 4 that is only 6 extra, not the 9 that doubling would need. Term 1 shows the claim failing at once: 3 and 3 are equal, not double.

9. Patterns P and Q as ordered pairs.

(a) (Open response — judge by reading.) Pattern P: 0, 3, 6, 9, 12. Pattern Q: 0, 5, 10, 15, 20. Paired up, P first: (0, 0), (3, 5), (6, 10), (9, 15), (12, 20). On the grid the five points lie on a straight line rising 5 for every 3 across.

(b) The pair whose first number is 6 is the third pair, so read its partner straight from the list: 6 is P’s third term and Q’s third term is 5×2 .

10

Either route is fine — reading the third pair, or noticing that each Q term is $\frac{5}{3}$ of its P term and computing $\frac{5}{3} \times 6$.

Look for: pairs in the right order (P first), points at the correct crossings, and an answer of 10 justified from the pattern or from the graph rather than guessed.

10. Write your own pair of rules. (Open response — judge by reading.)

Model answer: Rule 1 — start at 2 and add 3 each time: 2, 5, 8, 11, 14. Rule 2 — start at 8 and add 12 each time: 8, 20, 32, 44, 56. Every term of the second pattern is four times the matching term of the first: $8 = 4 \times 2$, $20 = 4 \times 5$, $32 = 4 \times 8$, and so on.

How the rules had to be chosen: the second pattern’s starting number must be four times the first pattern’s starting number, and its step must be four times the first pattern’s step. Multiply both parts of the rule by 4 and every term is multiplied by 4. Starting both patterns at 0 also works, and then only the step has to be quadrupled.

Look for: five terms listed for each pattern, the “four times” relationship holding at *every* term (not just the last one), and an explanation that names both the starting number and the step — a student who scales only the step will produce a pattern that fails at term 1, which is exactly the error problem 8 planted.